

QUALIFICATIONS

Ph.D. Chemical Engineering

(Aug. 2009 – May. 2014)

The University of Akron, Akron, OH, USA

(GPA: 3.5/4.0)

- Specialized in **Bioprocess, Fermentation, and Microbiologically Influenced Corrosion**
- Dissertation: Effects of Microbial Attachment and Biofilm Formation on Microbiologically Influenced Corrosion

B.E. Bioengineering

(Sep. 2003 – Jun. 2007)

Beijing Technology and Business University, Beijing, China

(GPA: 3.4/4.0, top 10% of class)

BACKGROUND

Bioprocess, Fermentation and Microbiologically Influenced Corrosion (MIC) Areas of Expertise:

- Five years hands-on research experience in bioprocess, fermentation, and microbiologically influenced corrosion
- Designed and constructed flow chamber/vessel systems for cell attachment studies to allow real-time monitoring of cell attachment under optical microscope
- Mastered microbiological techniques, e.g. aseptic technique, sterile filtration, organism identification, cultivation and storage of microorganisms
- Incubated level 1 and 2 microorganisms, such as iron-oxidizing bacteria: *A. ferrooxidans*, sulphate-reducing bacteria: *D. vulgaris*, and acid-producing fungi: *A. niger* and *T. reesei*.
- Conducted cell surface characterization by centrifuge and filtration
- Explored *P. aeruginosa* fermentation to produce *Rhamnolipid* under aerobic, microaerobic, and denitrifying conditions
- Studied *P. aeruginosa*, *E. coli*, and *P. putida* initial bacterial attachment onto substrates with different surface properties
- Investigated effects of *Rhamnolipid* as a biosurfactant and different shear stresses on initial attachment of *S. epidermidis*, *P. aeruginosa*, *E. coli* and *P. putida* in flow chamber system
- Examined the process of biofilm formation and biofouling initiated by bacterial attachment
- Conducted corrosion analysis, including corrosion caused by bacteria (iron-oxidizing and sulphate-reducing bacteria) and fungi (acid-producing fungi) on metal surfaces (e.g. Al alloy and carbon steel)

Surface Engineering and Material Science Areas of Expertise:

- Coated/modified and characterized hydrophobic or hydrophilic surfaces on metals/inorganic surfaces
- Determined surface tension of solid substrates, organic liquids and water based solutions
- Created nano-structure pattern by thermal responded dewetting of polymer thin films, and characterized by AFM and SEM
- Produced polymeric electrospun nano-fiber filters (super-hydrophilic/hydrophobic) for separation of liquid-liquid coalescence

Technical Skills:

- Optical/Fluorescence Microscopy, Atomic Force Microscopy (AFM), Scanning Electron Microscopy/Energy dispersive X-ray spectroscopy (SEM/EDS), Laser Scanning Confocal Microscopy (LSCM), Enzyme-linked immunosorbent assay (ELISA)
- Ellipsometry, Surface profiler, Sessile and Dynamic Liquid Contact Angle, Drop Shape Analyzer
- UV-VIS, Surface plasmon resonance (SPR), Dynamic light scattering (DLS), HPLC, GC and Extensive knowledge and hands-on experience in laboratory techniques

RESEARCH AND WORK EXPERIENCE

- Research Assistant, The University of Akron, Akron, OH, USA

(Aug. 2009-Present)

- Specialized in Bioprocess, Fermentation, and Microbiologically Influenced Corrosion

- **Research Assistant, Beijing Technology and Business University, Beijing, China** (Sep. 2007-Jun. 2009)
 - Separated and identified food flavor components by HPLC and GC-MS for food additives
 - Applied continuous counter current ultra-sonication to extract curcumin from turmeric
- **Research Assistant, Institution of Process Engineering, Chinese Academy of Science, Beijing, China** (Feb. 2007-Jun. 2007)
 - Electrospun glutenin with polymeric materials into nanofibers, applied for an **Invention Patent of China, Application Number: 200710121224**
- **Internship, Henan Lianhua Monosodium Glutamate Co., Ltd., Henan, China** (Oct. 2006-Jan. 2007)
 - Studied manufacturing process and step controls of Monosodium Glutamate formation
 - Researched fermentation vessel design and improved yield rates 3.6% by adjusting reaction temperature and pH during protein fermentation
- **Internship, Beijing Huiyuan Beverage & Food Group Co., Ltd., Beijing, China** (Jun. 2006-Sep. 2006)
 - Learned manufacturing process of fruit juices, had hands-on experience in Aseptic Cold Filling Lines
- **Internship, SINOPEC Beijing Yanshan Company, Beijing, China** (Jun. 2005-Sep. 2005)
 - Studied refining technologies, learned light oil, lubricant, and other oil production technology
 - Investigated low-grade crude process and increased light oil yield rates 2.1% by modifying condensation system
 - Learned production of monomer and polymer of synthetic fiber, synthetic resin, and rubber

PUBLICATIONS

- **H. Wang**, M. Sodagari, Y. Chen, X. He, B.-m. Zhang Newby*, L.-K. Ju*, “Initial bacterial attachment in slow flowing systems: effects of cell and substrate surface properties”, *Colloid. Surf. B: Biointerfaces*, 87 (2011) 415-422.
- **H. Wang**, M. Sodagari, Y. Chen, Q. Tang, X. Shan, G. Cheng, J. Payer, L.-K. Ju, B.-m. Zhang Newby*, “Developing flow system for monitoring initial stages of biofilm formation on microbiologically induced corrosion”, *Conference Proceeding, 2011 DoD Corrosion Conference* (Paper ID 20477).
- M. Sodagari, **H. Wang**, B.-m. Zhang Newby*, L.-K. Ju*, “Effects of rhamnolipids on initial attachment of bacteria on glass and octadecyltrichlorosilane-modified glass”, *Colloid. Surf. B: Biointerfaces*, 103 (2013) 121-128.
- **H. Wang**, M. Sodagari, L.-K. Ju, B.-m. Zhang Newby*, “Effects of shear on initial bacterial attachment in slow flowing systems”, *Colloid. Surf. B: Biointerfaces*, 109 (2013) 32-39.
- Q. Liu, W. Li, **H. Wang**, L. Liu*, “A Facile Method of Using Sulfobetaine-Containing Copolymers for Biofouling Resistance”, *J. Appl. Polym. Sci.* (in press)
- **H. Wang**, B.-m. Zhang Newby*, “Applicability of the XDLVO theory for interpreting adsorption of bovine serum albumin on solid surfaces”, (manuscript submitted to *Colloid. Surf. B: Biointerfaces*).
- **H. Wang**, L.-K. Ju, B.-m. Zhang Newby*, “Effects of *Pseudomonas aeruginosa* and *Escherichia coli* biofilm formation on microbiologically induced corrosion of carbon steel and aluminium alloy”, (manuscript in preparation).
- **H. Wang**, L.-K. Ju, B.-m. Zhang Newby*, “Corrosion behaviors of carbon steel and stainless steel in the presence of iron-oxidizing bacteria”, (manuscript in preparation).

VOLUNTEER SERVICE

- Mentored high school students and teachers in the High School Summer Corrosion Research Academy (NCERCAMP CRA)
- Mentored undergraduate students in Design Projects and Honors Projects

PROFESSIONAL AFFILIATIONS

- **Member**, American Chemical Society
- **Member**, NACE International – The Corrosion Society